International Institute of Information Technology, Hyderabad
MA4.101-Real Analysis (Monsoon-2026)

Assignment 3

Total marks: 100 Due date: 28 Nov, 2026

Instructions

1. There are seven questions, carrying the marks shown against each question and against each part.
2. You may use without proof any definition or result (proposition, lemma, theorem or corollary) stated in Chapters 5, 6 and 7 and Sections 9.1–9.3 of Tao's *Analysis I*, together with the fact stated in the box below. Any other result you wish to use must be proved.
3. Write complete arguments. A correct final answer with an unjustified step in the middle will not receive full marks, and a clearly written wrong turn will receive more than an unexplained correct one.
4. State at each step which definition or fact you are appealing to.
5. You may discuss the problems with others, but the solutions you submit must be written by you alone.

Fact you may assume

(F1) *(Division with remainder.)* For all integers $n \geq 0$ and $k \geq 1$ there are integers $q \geq 0$ and r with $0 \leq r < k$ such that $n = qk + r$.

Question (1) [14 marks]. Let $A, B \subseteq \mathbb{R}$ be non-empty and bounded above.

(a) **[9 marks]** Show that $A \cup B$ is bounded above and that

$$\sup(A \cup B) \;=\; \max\big(\sup(A),\ \sup(B)\big).$$

(b) **[5 marks]** Suppose in addition that $A \cap B \neq \emptyset$. Show that

$$\sup(A \cap B) \;\leq\; \min\big(\sup(A),\ \sup(B)\big),$$

and give an example in which this inequality is strict.

> **About this question.** Having seen how the supremum behaves under addition of sets, it is natural to ask how it behaves under the set operations themselves. The answer is lopsided. For unions there is an exact formula, because every element of either set is still present in the union. For intersections only an inequality survives, since the intersection may be a small part of each set, lying far below the top of either; two sets of two elements each are enough to see this.

Question (2) [15 marks]. Let a_1 and a_2 be any real numbers, and define

$$a_{n+2} \;:=\; \tfrac{1}{2}\big(a_{n+1} + a_n\big) \qquad (n \geq 1),$$

so that each term after the second is the average of the two before it.

(a) **[4 marks]** Show that $a_{n+2} - a_{n+1} = -\tfrac{1}{2}(a_{n+1} - a_n)$ for every $n \geq 1$, and deduce that $|a_{n+1} - a_n| = |a_2 - a_1|/2^{\,n-1}$.

(b) **[4 marks]** Show that the quantity $a_n + 2a_{n+1}$ takes the same value for every $n \geq 1$.

(c) **[7 marks]** Deduce that (a_n) converges, and that its limit is $\tfrac{1}{3}(a_1 + 2a_2)$. Verify this against the terms a_3, a_4, a_5, a_6 in the case $a_1 = 0$, $a_2 = 1$.

> **About this question.** Two separate things have to be established about a recursively defined sequence, and this question keeps them apart: part (a) controls how fast the sequence settles down, and part (b) controls where it settles. The quantity in (b) is a *conserved quantity*. The rule changes every term, yet leaves this one combination untouched, so the limit is forced to satisfy a single linear equation. Looking for something that a rule does not change is one of the most useful habits in the subject. Afterwards, try to find the conserved quantity for $a_{n+2} := \tfrac{1}{3}a_{n+1} + \tfrac{2}{3}a_n$.

Question (3) [12 marks]. Let $(a_n)_{n=1}^{\infty}$ be a sequence of reals which is bounded below, and let $(\varepsilon_n)_{n=1}^{\infty}$ be a sequence of non-negative reals for which $\sum_{n=1}^{\infty} \varepsilon_n$ converges. Suppose that

$$a_{n+1} \;\leq\; a_n + \varepsilon_n \qquad \text{for every } n \geq 1.$$

(a) **[7 marks]** Put $b_1 := a_1$ and $b_n := a_n - \sum_{k=1}^{n-1} \varepsilon_k$ for $n \geq 2$. Show that (b_n) is decreasing and bounded below.

(b) **[5 marks]** Deduce that (a_n) converges.

About this question. A decreasing sequence which is bounded below converges. This question asks how much of that survives if the sequence is allowed to go *up* a little at each step. If the allowances add up to something finite, subtracting what has been used so far leaves an honestly decreasing sequence, and everything survives. If they merely tend to 0, the conclusion can fail: allowances of $1/n$ add up to infinity, and a sequence using them can climb from 0 to 1 and fall back again forever. As with series, what matters is not whether small quantities are small, but whether they add up.

Question (4) [17 marks]. A sequence $(a_n)_{n=1}^{\infty}$ of reals is called *subadditive* if

$$a_{m+n} \;\le\; a_m + a_n \qquad \text{for all integers } m, n \ge 1.$$

Let $(a_n)_{n=1}^{\infty}$ be subadditive, and suppose that the set $\{\, a_n/n : n \ge 1 \,\}$ is bounded below, so that $L := \inf\{\, a_n/n : n \ge 1 \,\}$ is a real number.

(a) **[3 marks]** Show that $a_{qk} \le q\, a_k$ for all integers $q, k \ge 1$.

(b) **[7 marks]** Fix an integer $k \ge 1$. Show that there is a real number $C_k \ge 0$, depending on k but not on n, such that

$$\frac{a_n}{n} \;\le\; \frac{a_k}{k} + \frac{C_k}{n} \qquad \text{for every } n \ge k.$$

(Hint. Write $n = qk + r$ as in (F1)*, and treat the case $r = 0$ separately.)*

(c) **[7 marks]** Deduce that $(a_n/n)_{n=1}^{\infty}$ converges, and that $\lim_{n\to\infty} a_n/n = L$. *(This is Fekete's lemma.)*

About this question. Subadditive sequences turn up whenever a task of size $m+n$ can be done as a task of size m followed by one of size n. Fekete's lemma says that the cost per unit, a_n/n, always settles down, and that in the long run it is as good as the best value ever achieved, even though nothing in the hypothesis says that the sequence improves as n grows. The sequence (a_n/n) need not be monotone, so monotone convergence is of no use. Division with remainder takes its place: cut a task of size n into blocks of size k and a short leftover piece, whose cost is bounded and disappears after dividing by n.

Question (5) [13 marks]. Let $(a_n)_{n=1}^{\infty}$ be non-negative and decreasing, and suppose that $\sum_{n=1}^{\infty} a_n$ converges.

(a) **[9 marks]** Show that for every $n \geq 1$,

$$n\,a_{2n} \;\leq\; \sum_{k=n+1}^{2n} a_k \qquad \text{and} \qquad (n+1)\,a_{2n+1} \;\leq\; \sum_{k=n+1}^{2n+1} a_k,$$

and deduce that $\lim_{n\to\infty} n\,a_n = 0$.

(b) **[4 marks]** Show that the hypothesis that (a_n) is decreasing cannot be dropped. Let $a_n := 1/n$ if n is a perfect square and $a_n := 1/n^2$ otherwise, and show that $\sum a_n$ converges but $(n\,a_n)$ does not tend to 0.

> **About this question.** The zero test says that the terms of a convergent series tend to 0. For decreasing terms this question says much more: they tend to 0 faster than $1/n$. This explains at a glance why the harmonic series diverges. The idea is that a block of n consecutive terms of a decreasing sequence is at least n times its last term, while far out in a convergent series every block has a small sum. Part (b) shows that monotonicity is essential, since a sequence that is not monotone can hide large terms at sparse places.

Question (6) [13 marks]. For series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ define

$$c_n \;:=\; \sum_{k=0}^{n} a_k\,b_{n-k} \;=\; a_0b_n + a_1b_{n-1} + \cdots + a_nb_0 \qquad (n \geq 0);$$

the series $\sum_{n=0}^{\infty} c_n$ is called the *Cauchy product* of the two series. Write A_N, B_N, C_N for the partial sums from $n = 0$ to $n = N$.

(a) **[8 marks]** Suppose that $a_n \geq 0$ and $b_n \geq 0$ for every n. Show that

$$C_N \;\leq\; A_NB_N \;\leq\; C_{2N} \qquad (N \geq 0).$$

(Hint. Write each of the three quantities as a sum of the numbers a_jb_k over a set of pairs (j,k).)

(b) **[5 marks]** Suppose in addition that $\sum a_n = A$ and $\sum b_n = B$ converge. Deduce that $\sum c_n$ converges, with sum AB.

> **About this question.** For finite sums one multiplies every term of the first by every term of the second and adds up all the products. The Cauchy product orders these infinitely many products by collecting the pairs (j,k) with $j+k=n$, which is exactly how polynomials and power series are multiplied. For non-negative terms it always works, because A_NB_N, a sum over a square of pairs, is trapped between sums over

two triangles. For series which converge only by cancellation the product can fail completely: with $a_n = b_n := (-1)^n/(n+1)^{1/2}$ its terms do not even tend to 0. (If one of the two series converges absolutely, the Cauchy product does converge to AB; this is a theorem of Mertens.)

Question (7) [16 marks]. Let $f : \mathbb{R} \to \mathbb{R}$ satisfy

$$f(x+y) \;=\; f(x) + f(y) \qquad \text{for all real } x, y,$$

and put $c := f(1)$.

(a) **[5 marks]** Show that $f(0) = 0$, that $f(-x) = -f(x)$, and that $f(rx) = rf(x)$ for every rational r and every real x. In particular $f(r) = cr$ for every rational r.

(b) **[6 marks]** Suppose now that $\lim_{x \to 0;\, x \in \mathbb{R}} f(x)$ exists. Show that this limit equals 0, and that

$$\lim_{x \to x_0;\, x \in \mathbb{R}} f(x) \;=\; f(x_0) \qquad \text{for every } x_0 \in \mathbb{R}.$$

(c) **[5 marks]** Deduce that, under the hypothesis of (b), $f(x) = cx$ for every real x.

About this question. Every function $f(x) = cx$ satisfies $f(x+y) = f(x) + f(y)$, and one would like these to be the only solutions. Part (a) shows that the rule alone pins f down on the rationals, by pure algebra, but it says nothing at the irrationals. The surprise is how little analysis closes the gap: a limit at the single point 0, which the additive rule then carries to every other point. Without some such hypothesis the conclusion is false, although constructing a counterexample needs the axiom of choice.